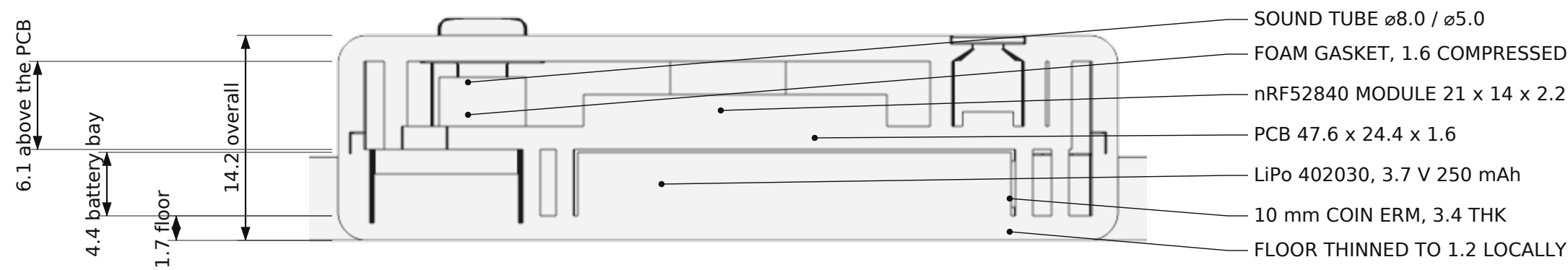
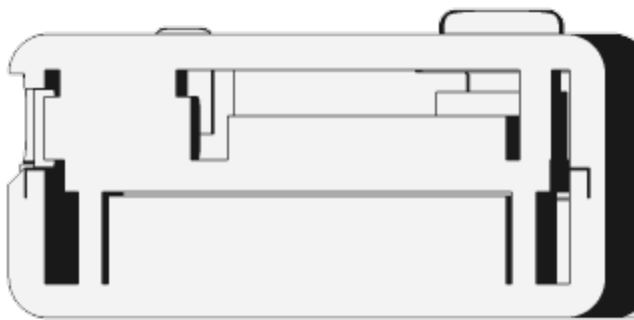


SECTION A-A (cut on Y = 0, viewed from +Y) scale 2.5:1

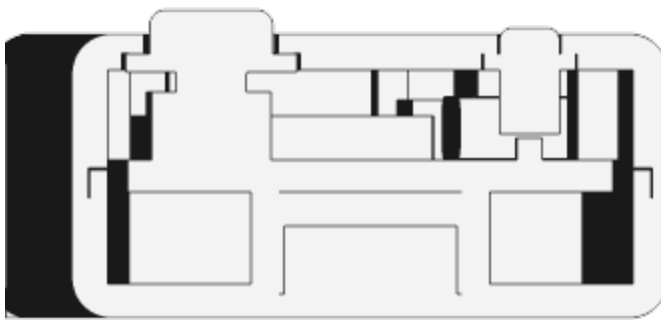


SECTION B-B (X = 0)



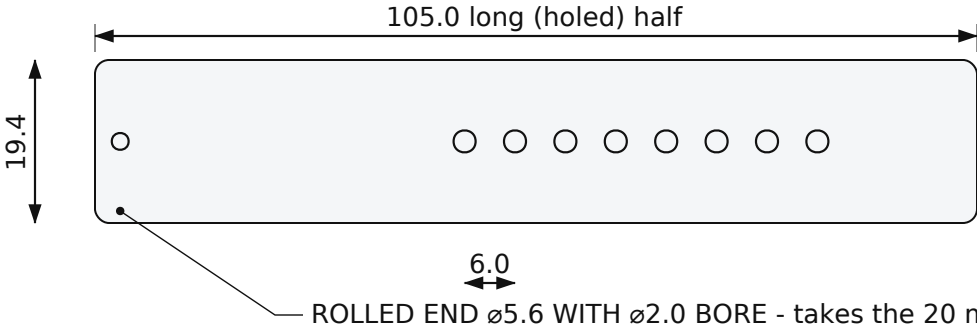
USB-C mating face sits flush with the inner wall - the plug shell is carried by the case, not by the solder joints.
0.20 clearance between the battery top and the PCB.

SECTION C-C (X = 17)



Cap flange $\varnothing 8.6$ is captive in the $\varnothing 9.0$ counterbore - the switch spring holds it up, so it cannot fall out or rattle.
Light pipe stands 0.3 proud; its collar is captive under the ceiling and it bottoms 0.2 above the LED.

STRAP DETAIL (developed / as printed flat)



Short (buckle) half 78.0 long, otherwise identical; it carries a second rolled end that captures the buckle bar.
Section 19.4 wide x 2.8 thick. 8 x $\varnothing 2.6$ lace holes at 6.0 pitch, first hole 44.0 from the bar axis.
Print flat in TPU 95A and heat-set to R27.0, or print pre-curved. Fits 150 - 200 mm wrists. Replaceable with any 20 mm watch strap.

VERTICAL STACK-UP (Z from the wrist face)

0.00 - 1.70	bottom shell floor
1.20 - 4.60	vibration motor (floor recessed)
1.70 - 6.10	battery + swelling allowance
6.00	PARTING LINE
6.30 - 7.90	PCB
7.90 - 11.06	USB-C receptacle
7.90 - 12.20	tactile switch
7.90 - 9.50	acoustic foam gasket
9.50 - 12.40	sound tube
12.40 - 14.20	top shell ceiling

RECOMMENDED ASSEMBLY ORDER

- 1 Press 4 x M2 heat-set inserts into the TOP shell bosses with a soldering iron at 200 C. Keep them square.
- 2 Glue the mesh disc into the mic recess (top face, outside).
- 3 Fit the light pipe AND the button cap from the INSIDE of the top shell - both flanges are larger than their bores, which is what keeps them captive. They cannot be fitted later.
- 4 Solder the motor and battery leads to the PCB while it is still out of the case. Leave 25 mm of slack on each.
- 5 Bond the motor into the bottom-shell pocket (thin CA or VHB).
- 6 Drop the battery into its fence; route the leads through the notch in the -X rib.
- 7 Seat the acoustic foam ring on the PCB around the mic.
- 8 Lower the PCB in - the 4 bosses enter its corner reliefs and locate it. It rests on the fence and the seat ribs.
- 9 Lower the TOP shell straight down; the tongue enters the groove and the clamp ribs press the PCB onto its seats.
- 10 Fit 4 x M2 x 8 CSK screws from the WRIST side. Cross-tighten to snug only - do not strip the inserts.
- 11 Fit the spring bars and both strap halves; add the keeper.
- 12 Test: press the button, charge over USB-C, tap near the mic.

PARTS AND FASTENERS

01	Bottom shell	1	PLA / PETG, printed
02	Top shell	1	PLA / PETG, printed
03	Button cap	1	PLA / PETG, printed
04	Light pipe	1	3 mm clear acrylic rod, 5.3 lg
05	Mic mesh disc	1	5.0 dia acoustic mesh
06	Mic gasket	1	8.0/5.0 foam ring, 2.0 thk
07	PCB assembly	1	47.6 x 24.4 x 1.6, 2-layer
08	MCU module	1	nRF52840 or ESP32-C3
09	Battery	1	LiPo 402030, 3.7 V 250 mAh
10	Vibration motor	1	10 mm coin ERM, 3.4 thk, 3 V
11	MEMS microphone	1	PDM top-port, SPH0645 class
12	Status LED	1	0805 RGB
13	Tactile switch	1	6 x 6 x 4.3 SMD
14	USB-C receptacle	1	16-pin SMD, charge + data
15	Env. sensor OPTION	1	BME688 / SHT40 / VEML7700
16	Spring bar	2	20 mm, 1.6 dia
17	Strap - buckle half	1	TPU 95A, printed
18	Strap - holed half	1	TPU 95A, printed
19	Buckle	1	PLA / PETG, printed
20	Keeper	1	TPU 95A, printed
21	Screw	4	M2 x 8 countersunk, steel
22	Heat-set insert	4	M2 brass, 4.0 lg, 3.5 OD

SENSE CLIP

TITLE	SECTIONS, INTERNAL LAYOUT, STRAP AND PARTS LIST
PROJECT	Sense Clip - wrist-worn assistive event sensor
MATERIAL	PLA / PETG (shells) TPU 95A (strap)
UNITS	millimetres PROJECTION third angle SCALE 2.5 : 1
TOLERANCE	general +/-0.30 fits as noted FDM as-printed
SHEET	2 of 2 REV A DATE 2026-08-19